

DocMind AI — Basic RAG Evaluation Report

Date: 2026-08-17 | **Chat model:** qwen2.5:0.5b (Ollama) | **Embedding model:** nomic-embed-text | **Corpus:** test_docs (sample.txt, sample.csv, sample.pptx)

1. Methodology

Baseline (Normal RAG): vector-only retrieval + llama-index default prompt & synthesizer.
Improved version: hybrid retrieval (vector + BM25 fusion) + explicit document-grounded prompt (TEXT_QA_TEMPLATE) + context budget trimming, via the same code path the app uses for chat.

20 questions were built from the three test documents: 6 text lookups, 6 CSV/tabular, 5 PPTX, and 3 negative (out-of-document) questions to test hallucination control. Both engines were run on identical questions with identical settings (top_k=3, temperature 0.4).

Scoring: Relevance 1–5 (does the answer address the question?) and Faithfulness 1–5 (are the facts grounded in the documents? 1 = hallucinated). No external evaluation library was used.

2. Results by Question

#	Type	Question	Expected	Baseline (vector-only)	Improved (hybrid)
1	txt	What is DocMind AI?	Offline/local RAG assistant	5/5 correct	5/5 correct
2	txt	Which chat models does DocMind use?	llama3, qwen2.5	2/4 answers "modes", not models	2/4 same error
3	txt	Which embedding model does DocMind use?	nomic-embed-text	3/4 lists it among models	2/4 vague
4	txt	What file types are supported?	TXT, CSV, DOCX, PPTX, PDF, MD	5/5 full list	1/3 rambles, no list
5	txt	Max documents per ingestion run?	300	5/5 300	5/5 300
6	txt	What license is DocMind released under?	MIT	1/1 "Apache 2.0" hallucinated	1/1 "Creative Commons" hallucinated

7	csv	SmartChip X9 Q4 revenue?	\$555,000	5/5 555,000	1/2 270,000 (wrong row)
8	csv	Alphabot 3000 units sold Q2 NA?	1350	5/5 1350	1/2 850 (wrong quarter)
9	csv	DocMind Pro total revenue NA?	\$705,000	2/4 refused (context had it)	1/3 247,500 (wrong row)
10	csv	Highest-revenue region for SmartChip?	Asia Pacific	1/2 North America	5/5 Asia Pacific
11	csv	SmartChip X9 Q1 revenue?	\$450,000	1/2 1200 (units)	1/2 3100 (units)
12	csv	Product with most total units?	SmartChip X9	1/4 refused	1/1 wrong (Alphabot)
13	pptx	Presentation title?	DocMind AI Launch Plan	5/5 correct	1/2 "Target Metrics"
14	pptx	The four launch milestones?	Q1–Q4 list	5/5 correct	5/5 correct
15	pptx	Target answer latency?	Under 5 seconds	5/5 correct	5/5 correct
16	pptx	Why is local RAG good for privacy?	Docs never leave device	4/5 correct	4/5 correct
17	pptx	Who prepared the presentation?	Product Team	2/4 refused	5/5 Product Team
18	neg	US President?	Refuse	1/1 "Joe Biden" hallucinated	5/5 refuses properly
19	neg	Capital of France?	Refuse	1/1 "Paris" hallucinated	1/1 "Paris" hallucinated
20	neg	Pasta carbonara recipe?	Refuse	1/1 full fabricated recipe	5/5 refuses properly

3. Summary

Metric	Baseline (Normal RAG)	Improved (Hybrid + Grounded Prompt)
Average relevance (1–5)	3.00	2.85
Average faithfulness (1–5)	3.65	3.50
Negative questions handled correctly	0 / 3	3 / 3
Correctly answered questions	10 / 20	9 / 20

4. Verdict

- **Headline win — hallucination control:** on out-of-document questions the improved version refused cleanly 3/3 times, while the baseline confidently invented facts (a fabricated president, a full fake recipe). This is the most valuable difference for real-world use.
- **Retrieval quality:** hybrid retrieval won where exact naming/keywords mattered (region question, "Product Team" slide).
- **Baseline strengths:** won 4 simple lookups (file types, Q4 revenue, units sold, presentation title) where vector search alone provided enough context.
- **Caveat:** the tiny 0.5B model limits both engines — aggregation and multi-row questions failed in both modes, and its noise partly masks retrieval differences. A re-run with llama3 would give a more decisive comparison.

5. Reproduction

Raw results: `scripts/eval_results.csv` | Runner: `scripts/evaluate.py` | Test corpus: `test_docs/` (sample.txt, sample.csv, sample.pptx)

Run with: `pipenv run python scripts/evaluate.py` (requires Ollama running with nomic-embed-text + qwen2.5:0.5b).